



QUESTION PAPER

Name of the Examination: WINTER 2023-2024 – CAT-1

Course Code: CSE 3004

Set number: 2

Duration: 90 min

Course Title: Design & Analysis of Algorithms

Date of Exam: 05/04/2024 (An) (A2)

Total Marks: 50

Q1. Apply suitable asymptotic notation to compute the time complexity of the following functions.

(5+5+5 = 15 Marks)

(i). $(n+a)^b = \Theta(n^b)$

(ii). Is $n! = o(2^n)$

(iii) Is $2^{n+1} = O(n^n)$

Q2. Solve the following recurrences relation using suitable method.

(5*2 = 10 Marks)

$$T(n) = 2T((n/2)+17) + n \text{ is } O(n \log n)$$

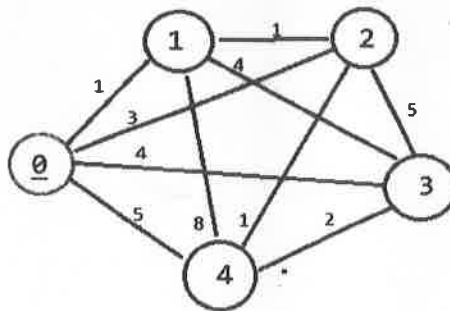
$$T(n) = 2T(n/4) + (\text{root}(n)) \quad \text{Make your bound as tight as possible.}$$

Q3. Given the following array [28, 31, 62, 34, 16, 58, 76, 97, 11, 84, 35, 40] and assume that quick sort will be used to sort this array in increasing order. Select the pivot value as convenience. Calculate the average-case complexity through recurrences relation. Also, explain why this makes quick sort perform efficiently.

(10 Marks)

Q4. Big basket agent daily drops the required foods to the address assigned in such a manner that he has to cover all the houses in the respective area with minimum travel cost. Compute the minimum travel cost using brute force method.

(5 Marks)



Q5. Apply merge sort algorithm to sort the list {Z, P, A, M, Y, M, F, S} in alphabetical order. Show all the steps in sorting. Is merge sort an in-place sorting algorithm? Justify? Also calculate the worst-case analysis of this algorithm through its recurrences function. (10 Marks)

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2,3			15
Q2	1	1	1,2,3			10
Q3	2	2	1,2,3			10
Q4	2	2	1,2,3			5
Q5	2	2	1,2,3			10

QUESTION PAPER

Name of the Examination: Winter 2023-24– CAT1

Course Code: CSE 3004

Set number: 3

Duration: 90 min

Course Title: Design and Analysis of Algorithms

Date of Exam: 06/02/2024 (FN) (B1)

Total Marks: 50

Q1. Solve the following recurrence relation and find the time complexity.

i) $T(n) = 7T(n/2) + 18n^2$,

ii) $T(n) = 2T(n/2) + n \log n$

(10M)

Q2. i) $f(n) = 2n^3 + 6n^2 + 5n + 7$. ii) $f(n) = 2^n + 3n^3 + n^2 + 5n$

Apply suitable asymptotic notations to compute the upper bound and lower bound time complexity of $f(n)$.

(10M)

Q3. Solve the recurrence relation using master method for $T(n) = 7T(n/2) + n$ and find the time complexity.

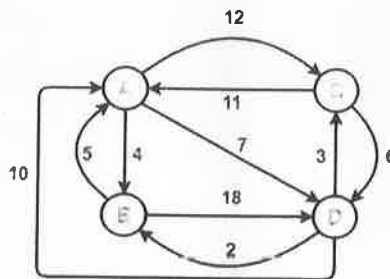
(5M)

Q4. Apply merge sort algorithm to sort the list { N, A, R, C, I, S, S, I, S, T, I, C } in alphabetical order. Show all the steps in sorting. Find the recurrence function and calculate the worst-case analysis of this algorithm.

(10M)

Q5. Sales agent daily required visit all the locations {A, B, C, D} in such a manner that he has to cover all the locations in the respective area with minimum travel cost. Compute the minimum travel cost using brute force method.

(10M)



Q6. In a shop, there are N items where each item has some weight and profit associated with it and also given a bag with capacity W , [i.e., the bag can hold at most W weight in it]. As customer, you collect the items into the bag such that the sum of profits associated with them is the maximum possible. Solve the following using brute force method. $N = 3$, $W = 5$, $\text{profit}[] = \{1, 2, 3\}$, $\text{weight}[] = \{4, 5, 1\}$

(5M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2,3	-	1	10
Q2	1	1	1,2,3	-	1	10
Q3	1	1	1,2,3	-	1	05
Q4	2	2	1,2,3	-	1	10
Q5	2	2	1,2,3	-	1	10
Q6	2	2	1,2,3	-	1	5